

Anoop Mishra

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EDUCATION

Mississippi State University

Graduation: May 2027 | GPA: 4.0 / 4.0

B.S. Computer Science | B.S. Mathematics

Relevant Coursework: Operating Systems, Data Structures & Algorithms, Computer Organization, Parallel Algorithms, Data Science, Data Communications & Networks, Machine Learning

WORK EXPERIENCE

Infrastructure Engineer Intern - Playbook | Athletics Intelligence Research Laboratory

Jun 2026 - Present

- Architected a multi-database PostgreSQL cluster storing 4.3M+ rows for 50+ users, with 40+ versioned migrations and per-service role isolation
- Implemented multi-tenant authorization with PostgreSQL Row-Level Security, enforcing default-deny access across 3 independent scopes
- Migrated production services to rootless Podman with systemd Quadlets on RHEL 9/SELinux, across 3 VMs and a dedicated GPU workstation
- Automated 3-environment CI/CD with GitHub Actions and GHCR, sequencing migrations before deployments to prevent version mismatches

Undergraduate Research Assistant | Wireless Communications Lab

Feb 2025 - Jul 2026

AI-Driven Network Automation (NTIA Funded)

- Built a self-healing multi-agent LLM orchestration system with LangChain and HuggingFace Transformers, reducing network configuration complexity by 70%
- Engineered a RAG pipeline backed by ChromaDB, improving O-RAN spec retrieval accuracy by 60%
- Benchmarked 5 open-source LLMs on custom efficiency metrics, surfacing a 6x performance gap to guide model selection

Open Fronthaul Security Testing (NSF Funded)

- Won IEEE MILCOM Outstanding Demo Award for co-authored research on Open Fronthaul protocol security testing
- Built an autonomous Bash testing engine with self-recovery, enabling 24/7 attack execution with zero human intervention
- Reduced test environment setup time by 75% via Linux network virtualization, Docker containers, and SDR UE emulation
- Streamed live logs to InfluxDB; built Grafana dashboards for real-time attack observability across test runs

AutoSec-RAN - 5G/O-RAN Security LLM Pipeline

- Designed a multi-step data generation system producing 4.8K+ security Q&A examples with validation passes for factual grounding
- Fine-tuned instruction-following LLMs with LoRA/PEFT on an A100 GPU using mixed precision, checkpointing, and memory-aware training
- Evaluated 168 security scenarios from network captures and logs, identifying 243 high/critical findings across DoS and injection-style attacks

AI Hub Fellow | Mississippi ITS AI Innovation Hub

Feb 2026 - May 2026

- Delivered a completed proof of concept and presented findings directly to Mississippi Board of Nursing
- Designed and implemented an AI/ML transcript fraud detection pipeline on AWS using Textract, Lambda, and Bedrock Knowledge Bases
- Built Lambda orchestration combining OCR extraction, prompt-based fraud checks, risk scoring, and explainable output generation
- Integrated the AI pipeline with a FastAPI backend via async invocation and S3 document flow with secure callback updates

Teaching Assistant | NSF ExLENT XAI Project

Feb 2026 - May 2026

- Graded and validated reference solutions across NLP and deep learning modules in an NSF-funded XAI curriculum

Data Science Assistant | Center for Advanced Vehicular Systems (CAVS)

Nov 2024 - Jul 2025

- Optimized construction resource allocation across multiple sites using mathematical models based on hardware constraints

PROJECTS

Synq - Cloud Native Systems Tracker | C++, PostgreSQL

Dec 2025 - Present

- Built a C++ monitoring agent with 0.3% CPU usage and 15 MB memory footprint, deployed via Docker on Supabase and Render

Wakapanel - Productivity Tracker | JavaScript (GJS), GTK4

Nov 2025 - Present

- Published a GNOME Shell extension integrating the WakaTime API to surface real-time coding stats on the desktop panel
- Deployed initial version; iterated on UI/UX based on user feedback, growing to 200+ users

TECHNICAL SKILLS

Languages: C/C++, Python, JavaScript (GJS), SQL, Bash, HTML/CSS

Infrastructure & DevOps: Podman, Docker, systemd, RHEL/SELinux, nginx, gunicorn, GitHub Actions, GHCR, CI/CD, Git, Linux

Frameworks & Libraries: PyTorch, HuggingFace, LangChain, FastAPI, Flask, GTK4, Scikit-learn, Pandas, NumPy, ChromaDB, FAISS

Cloud & Databases: PostgreSQL, InfluxDB, Grafana, Redis, Supabase, AWS (Lambda, S3, Textract, Bedrock, IAM)